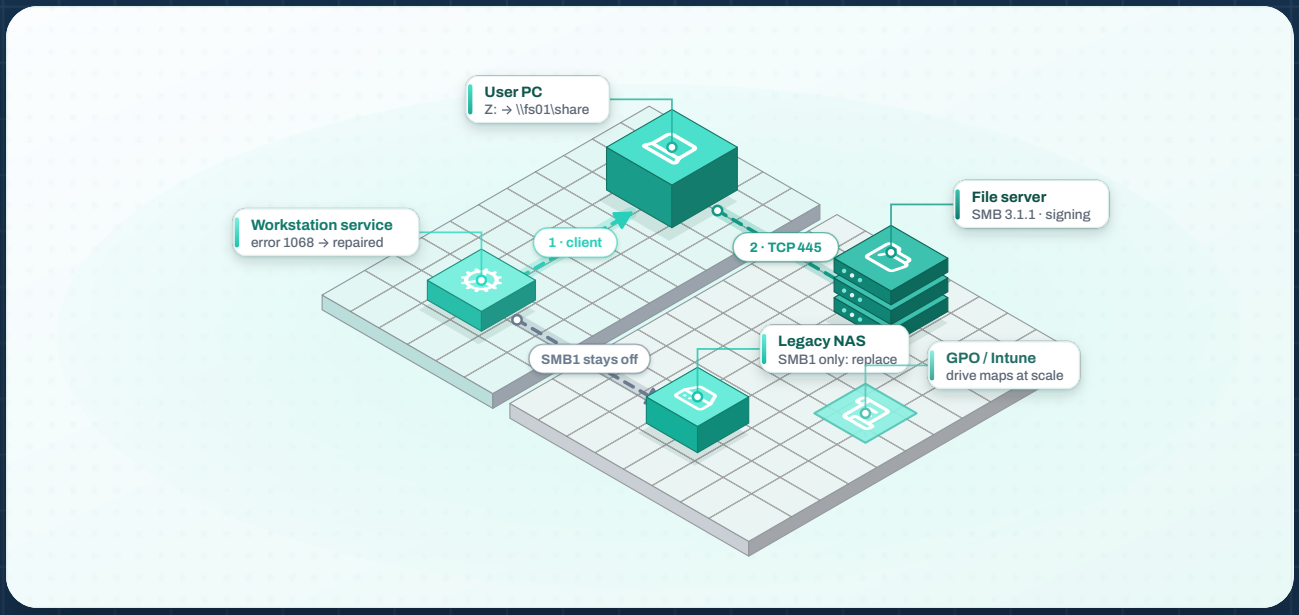
STANDARD OPERATING PROCEDURE · FIELD GUIDE SERIES

SOP-WIN-003

Restoring Mapped Network Drive Access on Windows 11

Recover the Workstation service and SMB client after a configuration change, then re-map drives four ways: Services and Windows Features, sc/net use/cmdkey, PowerShell SMB cmdlets, and Group Policy or Intune at scale.

Windows Desktop Support | File Sharing (SMB) | Endpoint Security



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Sanitized reference edition — client identifiers removed

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LEARNING OBJECTIVES

After working through this procedure you will be able to:

- Explain how Windows reaches a file share: the SMB redirector stack, the Workstation (LanmanWorkstation) service and its dependencies.
- Describe the SMB dialects, why SMB1 is insecure, and why it should stay disabled whenever the server supports SMB 2/3.
- Diagnose error 1068 on the Workstation service and repair it, preferring a dependency fix over re-enabling SMB1.
- Map, inspect and remove drives with net use, cmdkey and the SmbShare PowerShell cmdlets.
- Deploy drive mappings centrally with Group Policy Preferences or an Intune script.
- Troubleshoot the common net use errors, including Windows 11 24H2 signing and guest-access changes.

TIME	DIFFICULTY	MINIMUM ACCESS	RISK
15–40 min (two to three restarts)	Beginner to intermediate	Local administrator (service/feature changes)	Low; moderate if SMB1 is re-enabled

CHAPTER 1

Concepts You Need First

A **mapped network drive** is a drive letter (for example `Z:`) that points to a folder shared by another computer, written as a UNC path such as `\\fs01\share`. When mapped drives disappear, users usually see Explorer's "Windows cannot access \\server\share" dialog or a red X on the drive. The cause can be the network, the server, permissions, credentials, or, as in the reference engagement for this SOP, the *client* itself: a configuration-management package altered the workstation's SMB settings, the **Workstation** service could no longer start, and every mapped drive on the PC failed at once.

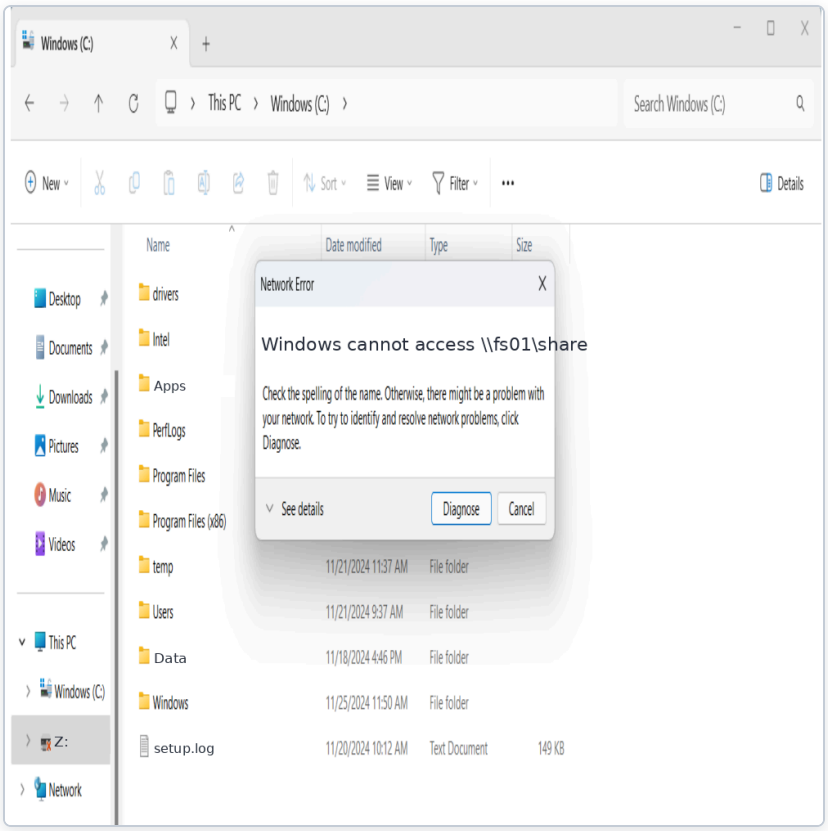


Figure 1. The symptom: Explorer cannot reach the share and the mapped drive in the navigation pane shows as disconnected. (Share and folder names replaced.)

1.1 How Windows reaches a file share

Windows talks to file servers with the **Server Message Block** (SMB) protocol over TCP port 445. On the client, SMB is implemented by a *redirector*: kernel drivers that make a remote share look like a local file system, so any program can open `Z:\report.xlsx` without knowing it is on another machine.

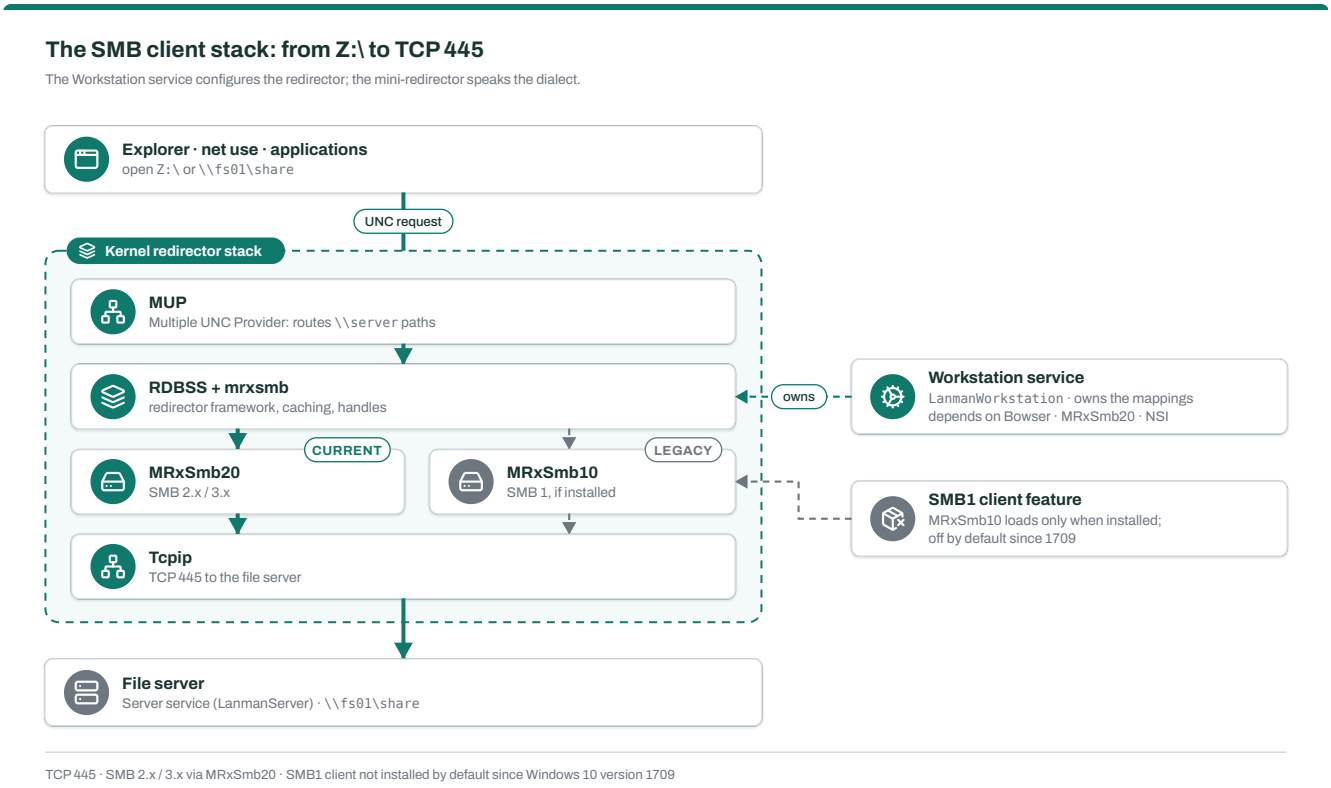


Figure 2. Mapped-drive faults on a user's PC are usually in the client half of this picture, which the Workstation service controls.

Component	Role
MUP	*Multiple UNC Provider*: receives every `\\server\share` request and hands it to the right network provider (SMB, WebDAV, NFS).
RDBSS / mrxsmb	Redirected Drive Buffering Subsystem and the common SMB redirector: caching, file handles, connection management.
MRxSmb20	Mini-redirector for SMB 2.x and 3.x. This is what all modern connections use.
MRxSmb10	Mini-redirector for SMB 1. Present only if the *SMB 1.0/CIFS Client* optional feature is installed.
Bowser	The legacy Computer Browser support driver (`bowser.sys`), still a dependency of the Workstation service.
Workstation service	Service name `LanmanWorkstation`. "Creates and maintains client network connections to remote servers using the SMB protocol. If this service is stopped, these connections will be unavailable."

KEY CONCEPT · CLIENT AND SERVER ARE DIFFERENT SERVICES

The **Workstation** service (LanmanWorkstation) is the SMB *client*: it lets this PC use other computers' shares. The **Server** service (LanmanServer) is the SMB *server*: it lets other computers use this PC's shares. Mapped-drive problems on a user's PC are almost always about Workstation; the server side is on the file server.

1.2 SMB versions and why SMB1 must stay off

When a client connects, client and server negotiate the highest **dialect** both support. Every dialect since 2006 is a large improvement on the original:

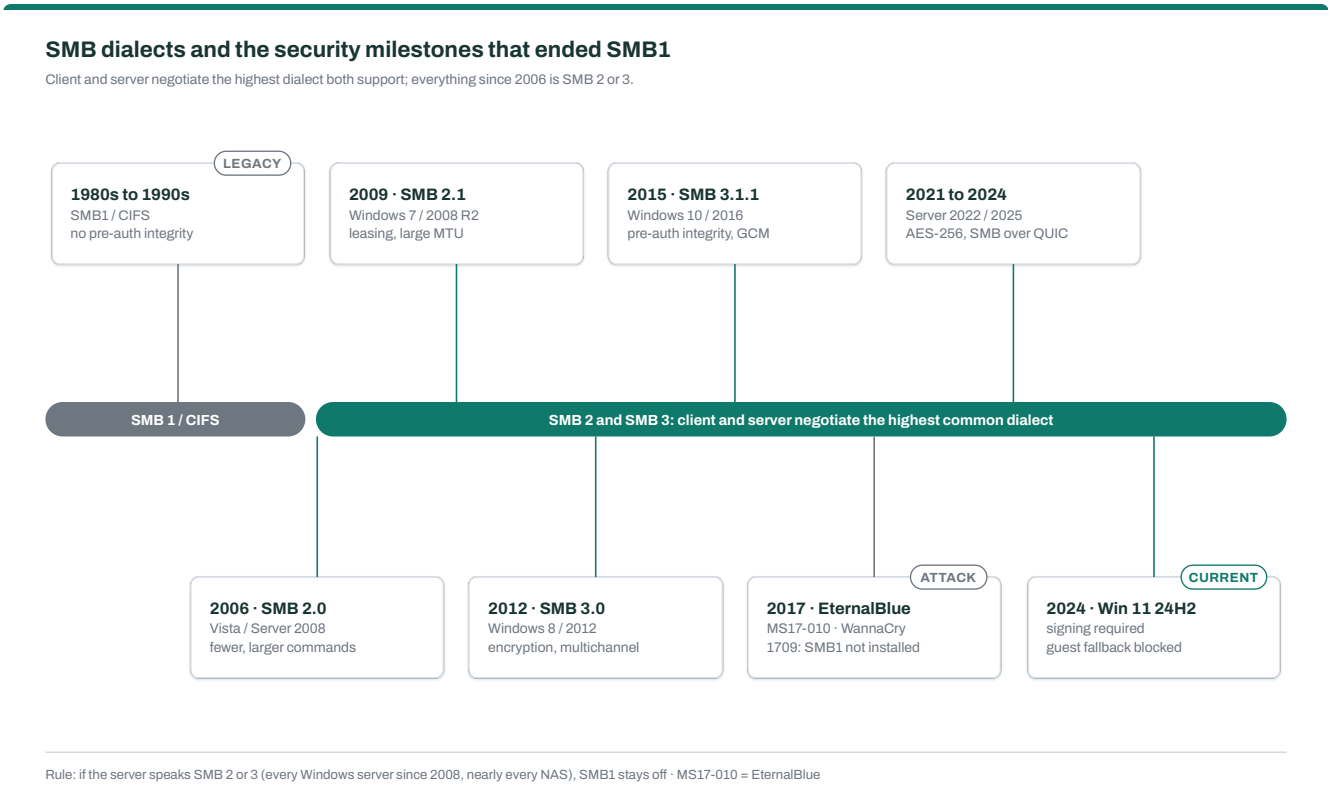


Figure 3. Every Windows server since 2008 speaks SMB 2 or 3, so a modern client never needs SMB1 to reach it.

SMB1 dates from the 1980s. It has no pre-authentication integrity (a man-in-the-middle can downgrade or tamper with the session), no encryption, weak signing performance and a large, old code base. In 2017 the *EternalBlue* exploit for an SMB1 flaw (MS17-010) powered the WannaCry and NotPetya outbreaks, which spread worm-like between unpatched machines. Microsoft has not installed SMB1 by default since Windows 10 version 1709, and the optional feature includes an *Automatic Removal* component that uninstalls the SMB1 client when it has not been used for a while.

RULE · SMB1 IS A LAST RESORT, NEVER A DEFAULT FIX

Re-enable SMB1 only when a specific, identified device (an old NAS, scanner or embedded system) supports *nothing newer*, only on the machines that must talk to it, and with an end date for replacing that device. If the server supports SMB 2 or 3, which every Windows server since 2008 and almost every NAS does, SMB1 must stay off.

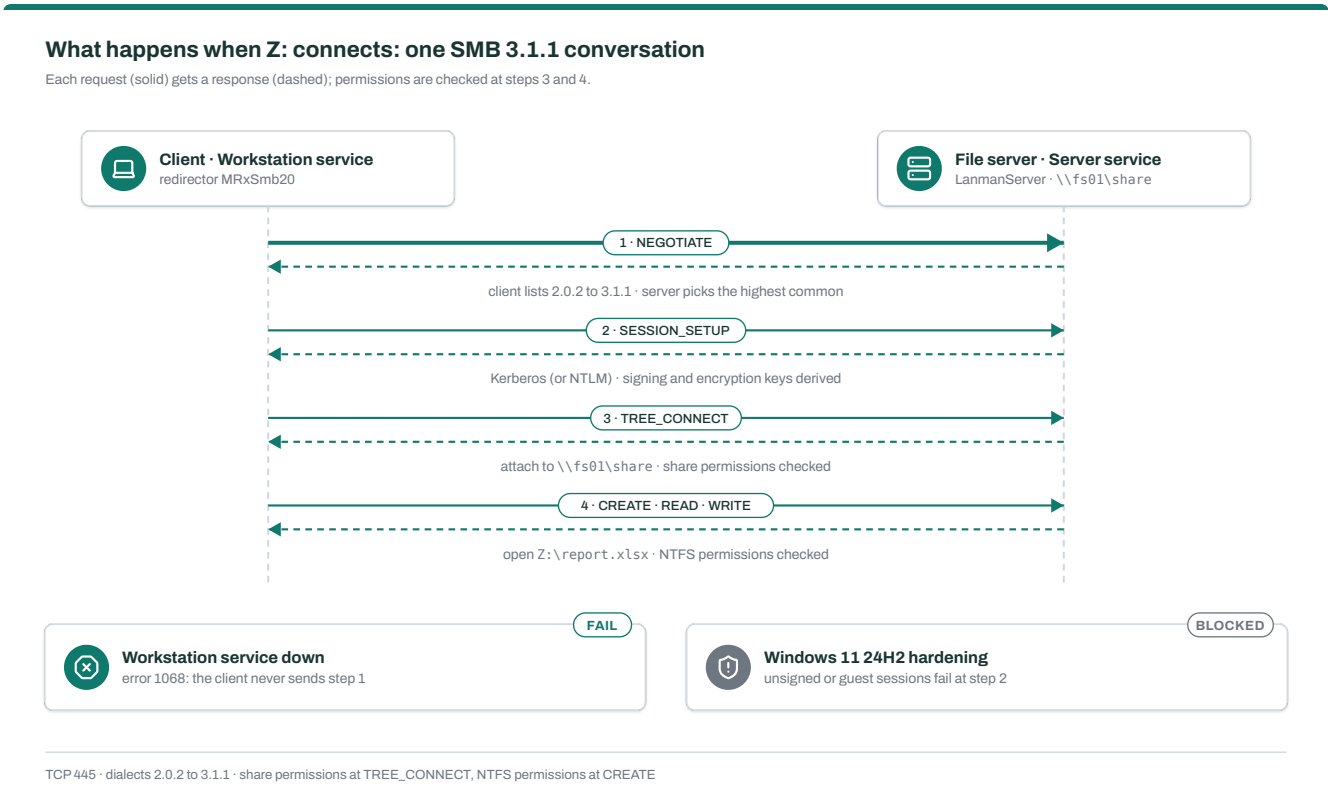


Figure 4. When the Workstation service is down none of this conversation happens, because the client never sends the first message.

1.3 The Workstation service and error 1068

Like every service, Workstation lists its **dependencies**, components the Service Control Manager must start first. On a current Windows 11 machine without the SMB1 client, the list is `Bowser`, `MRxSmb20` and `NSI`. When the SMB1 client is installed, `MRxSmb10` is added. If any listed dependency is disabled or missing, Workstation refuses to start with **error 1068: The dependency service or group failed to start**, and no SMB connection can be made.

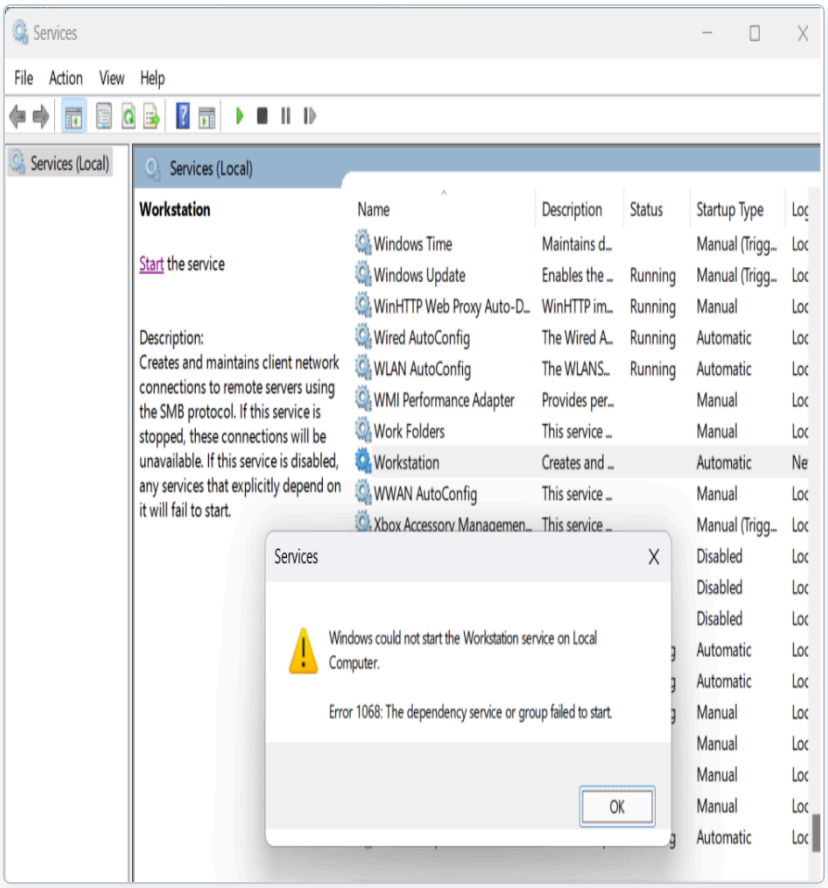


Figure 5. Starting the Workstation service from services.msc fails with error 1068. The description pane explains the impact: connections to remote servers "will be unavailable".

The most likely mechanism in the reference engagement, consistent with every symptom and with the fix that worked, is shown below: the package disabled or removed the SMB1 mini-redirector but left the Workstation service's dependency list still naming it.

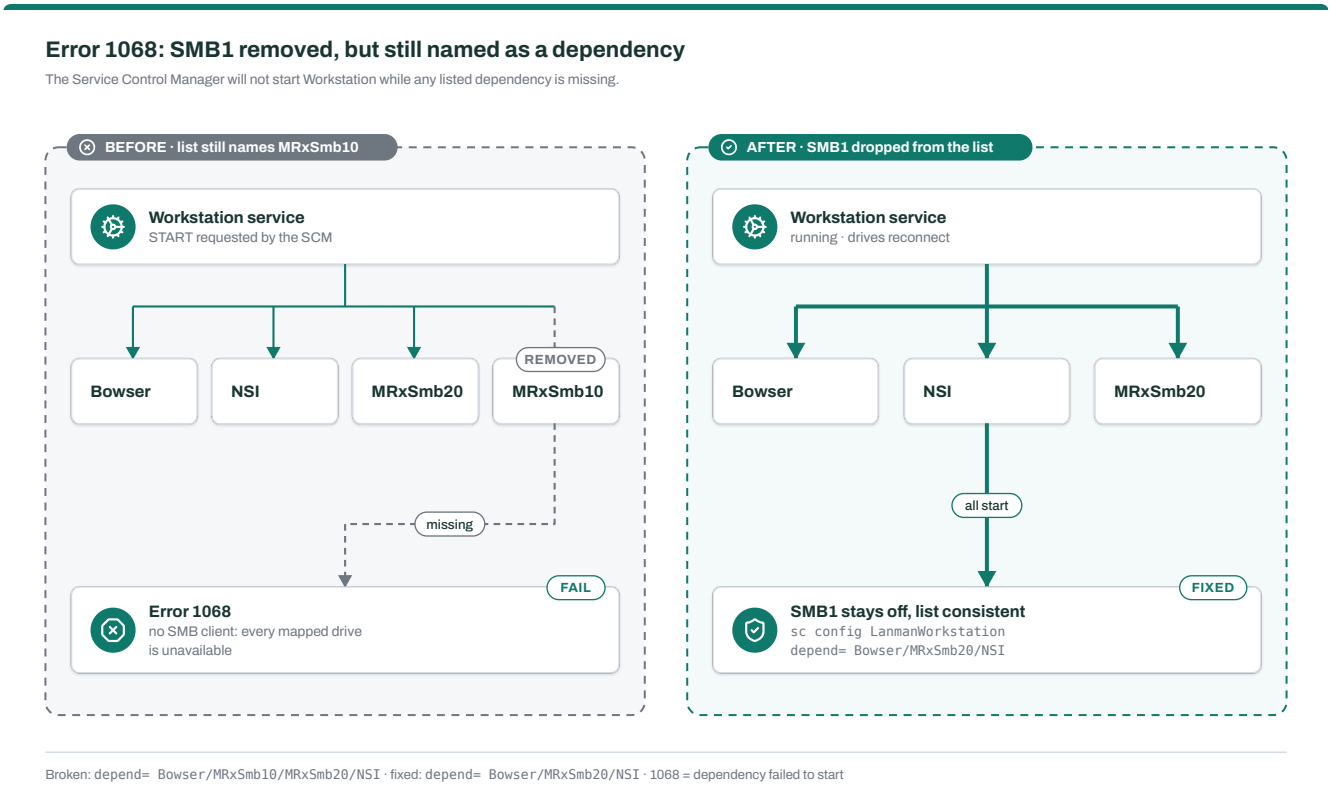


Figure 6. Removing SMB1 is correct; leaving it in the dependency list is what breaks the client.

WHY · TOGGLING THE SMB1 FEATURE FIXED IT

The original procedure uninstalled the package, turned *SMB 1.0/CIFS File Sharing Support* off, rebooted, turned it on again and rebooted. Turning the feature *off* through Windows Features makes Windows rewrite the Workstation dependency list consistently (removing MRxSmb10); turning it back on installs MRxSmb10 properly and adds it back. Either state is internally consistent, so Workstation starts. That means the "off" half of the procedure is the part that repairs the client, and the "on" half only re-introduces SMB1. This SOP therefore stops at "off" unless a legacy server truly needs SMB1.

1.4 Drive mappings, credentials and sessions

- **Persistent mappings** are stored per user (`HKCU\Network\<letter>`) and reconnected at logon. Mappings made by Group Policy or logon scripts are re-created at each logon.
- **Credentials**: domain users authenticate with Kerberos automatically. For a server outside the domain, saved credentials live in **Credential Manager** (Control Panel › Credential Manager › Windows Credentials, or `cmdkey`).
- **One credential per server per session**: Windows allows only one set of credentials to a given server name at a time. Mapping a second share on the same server with different credentials fails with error 1219.

- **Elevated versus standard tokens**: with UAC, an administrator has two logon tokens. Drives mapped in an elevated prompt are invisible in Explorer (standard token) and vice versa. Map drives from a normal, non-elevated session.

1.5 Choosing a method

A · GUI

services.msc, Programs and Features, Windows Features, Map Network Drive.

BEST FOR: ONE PC, HANDS-ON

B · Command line

sc, net use, cmdkey, DISM feature commands.

BEST FOR: FAST, EXACT REPAIR

C · PowerShell

SmbShare cmdlets, optional-feature cmdlets and a repair template.

BEST FOR: REPEATABLE, REMOTE

D · GPO / Intune

Drive Maps preferences, Intune scripts and remediations.

BEST FOR: MANY USERS, PREVENTION

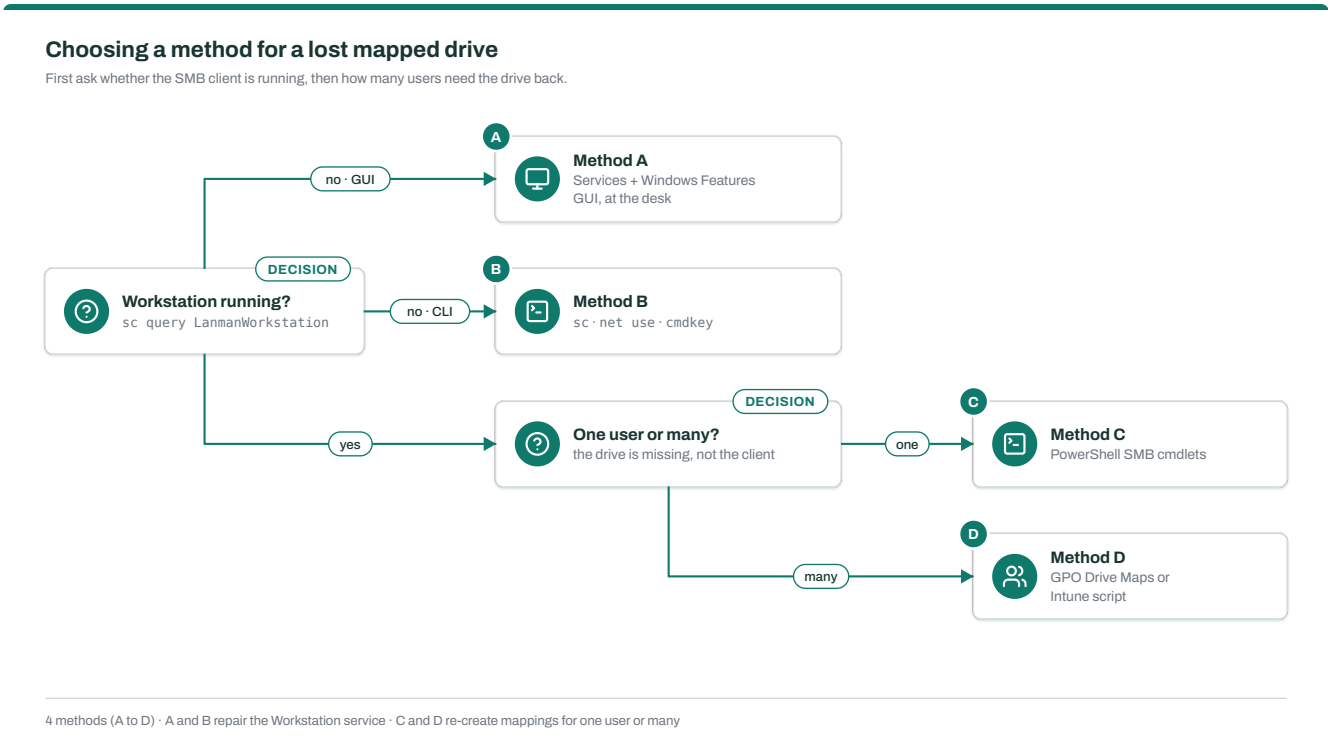


Figure 7. Repair the service first if it is stopped; only re-create mappings once the client itself is healthy.

CHAPTER 2

Access, Prerequisites and Planning

2.1 Level of access required

Task	Minimum access	Notes
Map or remove a drive for yourself	Standard user + share and NTFS permissions on the server	Do it in a non-elevated session so Explorer sees it.
View services, dependencies, SMB connections	Standard user	`sc qc`, `Get-SmbConnection` and `net use` work unelevated.
Start/configure services, change dependencies	Local Administrators, elevated	Needed for `sc config` and services.msc changes.
Uninstall software, change Windows optional features	Local Administrators	Includes SMB1 and the offending package. A managed package may reinstall itself: coordinate with its owner.
Change share or NTFS permissions	File server administrator / data owner	Not a desktop task; raise to the file-services team.
Group Policy Drive Maps	Edit rights on the user GPO, link rights on the OU	Delegated GPO editors.
Intune scripts and Remediations	Intune Administrator or custom role with *Device configurations* permissions	Remediations need Windows Enterprise/Education E3/E5 (or equivalent) licensing.

> **GOOD PRACTICE · MANAGED LOCAL ADMIN, AND TALK TO THE PACKAGE OWNER**
>
> Use the device's Windows LAPS password for elevation. Because the root cause was a centrally deployed configuration package, also notify the configuration-management owner: if the package is still assigned, it will reinstall and break the PC again, and other PCs are probably affected.

2.2 Prerequisites checklist

- [ ] Ticket number; list of the user's drive letters and UNC paths (from the user, the GPO or `net use` on a working PC).
- [ ] Confirmation the file server is up and reachable from another PC.

- [ ] Local administrator credential; user present to sign in after restarts.
- [ ] The name of any recently deployed package that changed SMB or network settings (software inventory, deployment history).
- [ ] Whether any device the user needs genuinely requires SMB1 (ask the file-services team; assume "no").

2.3 Risk and planning notes

Action	Risk	Notes
Inspect service, `net use`, `Get-SmbConnection`	READ-ONLY	Always first.
Uninstall the offending package	LOW	Check it is not required by other software.
Correct Workstation dependencies	LOW	Restores the shipped configuration for an SMB1-free PC.
Disable the SMB1 feature	LOW	Security improvement; requires a restart.
Enable the SMB1 feature	HIGH	Re-introduces a known-exploited protocol. Exception approval, scoped and time-boxed.

CHAPTER 3

Method A · Restoring Access Through the GUI

Work down the fix ladder below. Every method in this SOP follows the same order; only the tools change.

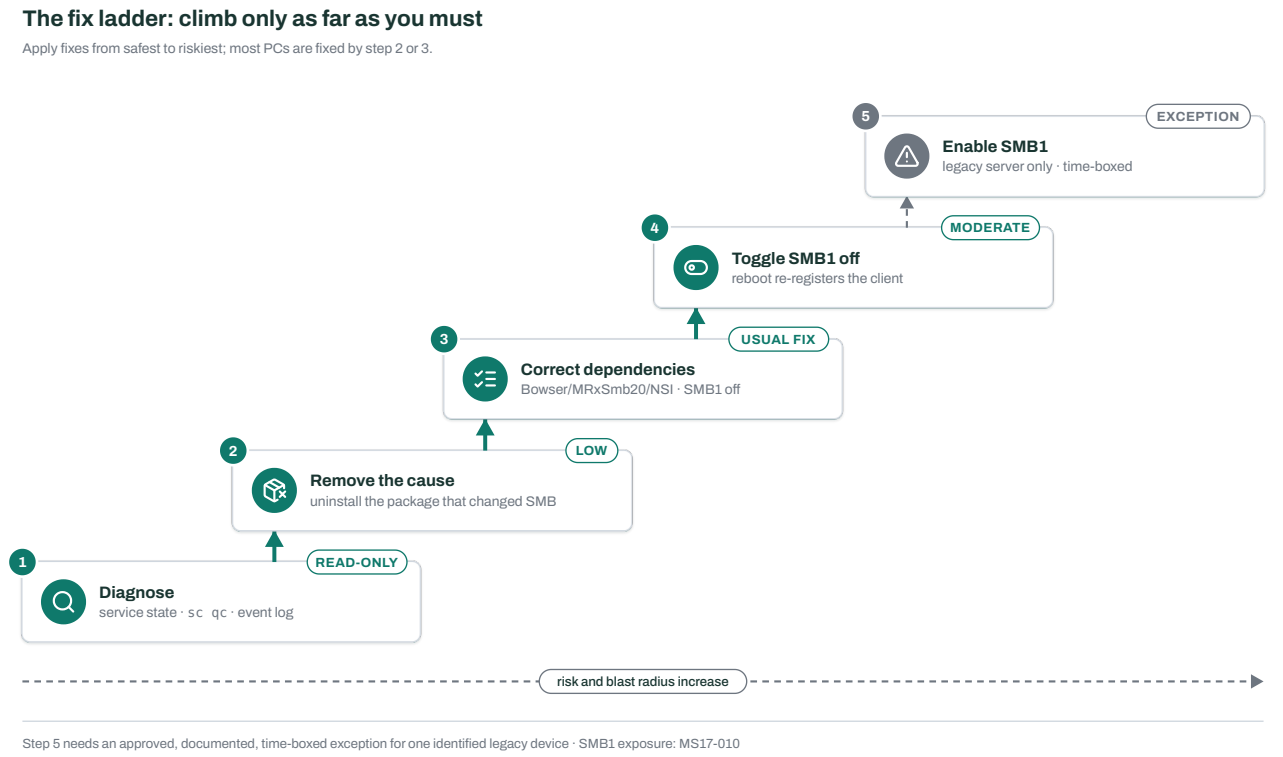


Figure 8. Stop at the lowest rung that works, because each step up widens the change and its security exposure.

STEP 1 Check the Workstation service

1. Press Win+R, type `services.msc` and select **OK**.
2. Find **Workstation**. If **Status** is *Running*, the client is healthy: go to Step 5 and re-map the drive.
3. If it is stopped, right-click › **Start**. If it starts, check why it was stopped (System event log) and go to Step 5.
4. If Windows reports **error 1068**, open **Properties › Dependencies**. Note every entry in the top box and check whether one of them (typically *SMB 1.x MiniRedirector*) is disabled or missing.

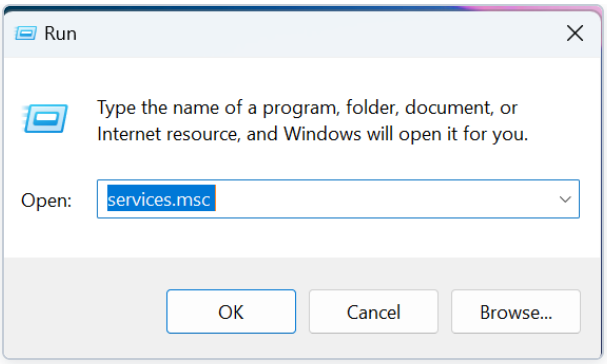


Figure 9. Opening the Services console from the Run dialog.

STEP 2 Remove the configuration package that changed SMB

1. Open **Settings › Apps › Installed apps** (or **Control Panel › Programs and Features**).
2. Locate the configuration-management package identified in planning (in the reference case, a small script package that altered SMB settings) and select **Uninstall**.
3. Restart the computer, then check the Workstation service again. Some packages restore what they changed on uninstall; many do not.

STEP 3 Repair the client with SMB1 left off (preferred)

1. Press Win + S, type **Turn Windows features on or off** and press Enter.
2. Scroll to **SMB 1.0/CIFS File Sharing Support** and expand it.
3. If the box is checked or partly filled, **clear** it and all its sub-options (Automatic Removal, Client, Server), select **OK**, and restart when prompted.
4. If it is already clear, the dependency list must be corrected directly: use Method B, Step 2 (one `sc config` command).
5. After the restart, confirm Workstation is *Running* with startup type *Automatic*.

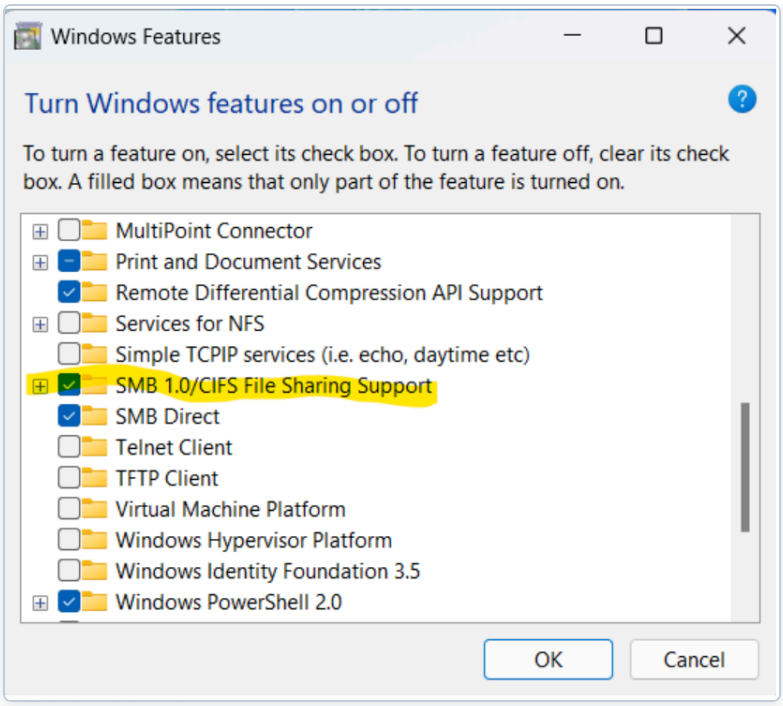


Figure 10. The SMB 1.0/CIFS File Sharing Support feature in Windows Features. Clearing it is the repair; leaving it cleared is the secure end state.

STEP 4 Only if a legacy server requires it: re-enable SMB1

This reproduces the second half of the original procedure. Do it only after confirming that a specific device needs SMB1 and an exception is approved.

1. Reopen `Turn Windows features on or off`, check `SMB 1.0/CIFS Client` only (not the Server sub-feature unless this PC must *serve* SMB1), select `OK` and restart.
2. Record the device that needs SMB1, the exception reference and the date by which it will be replaced.

WATCH OUT · AUTOMATIC REMOVAL

If *SMB 1.0/CIFS Automatic Removal* is checked, Windows removes the SMB1 client again after a period of non-use, which can bring back error 1068 on PCs with a stale dependency list. That is another reason to fix the dependency list rather than rely on the feature state.

STEP 5 Re-map and verify the drive

1. Sign in as the user (not elevated). Press `Win`+`E` and check whether the drives reconnected.
2. If not, select **This PC** › **...** › **Map network drive**.
3. Choose the drive letter, enter the UNC path (for example `\\fs01.contoso.com\share`), keep **Reconnect at sign-in** checked. Tick **Connect using different credentials** only for servers outside the user's domain.
4. Open a file on the drive and save a test file (if the user has write rights), then delete it.

CHAPTER 4

Method B · Command Line (sc, net use, cmdkey)

4.1 Step 1 · Inspect

COMMAND PROMPT · INSPECT (NO ELEVATION NEEDED) CMD

```
sc query LanmanWorkstation
sc qc LanmanWorkstation
sc query MRxSmb20
sc query MRxSmb10          &:: "does not exist" is normal on an SMB1-free PC
sc query Bowser
net use                    &:: current mappings and their status
```

On a healthy, SMB1-free Windows 11 PC the dependency lines of `sc qc LanmanWorkstation` read:

EXPECTED DEPENDENCIES (SMB1 CLIENT NOT INSTALLED) TEXT

```
        DEPENDENCIES       : Bowser
                           : MRxSmb20
                           : NSI
```

If `MRxSmb10` appears in that list while `sc query MRxSmb10` reports it missing or `sc qc MRxSmb10` shows `START_TYPE : 4 DISABLED`, you have found the cause of error 1068.

4.2 Step 2 · Repair (elevated)

ELEVATED COMMAND PROMPT · REPAIR WITHOUT SMB1 CMD

```
:: Make the dependency list match an SMB1-free client (space after "depend=" is required)
sc config LanmanWorkstation depend= Bowser/MRxSmb20/NSI
sc config LanmanWorkstation start= auto
sc start LanmanWorkstation

:: Make sure the SMB1 feature is off (removes MRxSmb10 cleanly; restart afterwards)
DISM /Online /Get-Features /Format:Table | findstr /i SMB1
DISM /Online /Disable-Feature /FeatureName:SMB1Protocol /NoRestart
```

NOTE · THESE ARE MICROSOFT'S DOCUMENTED VALUES

Microsoft's guidance for disabling the SMB1 client on older Windows versions uses exactly `sc.exe config lanmanworkstation depend= bowser/mrxsmb20/nsi` together with disabling `mrxsmb10`. The command above applies the same end state. Confirm the list against a healthy PC of the same build before scripting it.

If (and only if) an approved exception requires SMB1, the matching state is:

EXCEPTION ONLY · SMB1 CLIENT ENABLED CMD

```
DISM /Online /Enable-Feature /FeatureName:SMB1Protocol-Client /All
:: after restart, Windows sets the dependency list to Bowser/MRxSmb10/MRxSmb20/NSI
```

4.3 Step 3 · Re-map drives with net use

Run these in a normal (non-elevated) prompt as the user.

COMMAND PROMPT · MAPPINGS CMD

```
net use Z: \\fs01.contoso.com\share /persistent:yes
net use                                          &:: Status should be OK
net use Z: /delete                               &:: remove one mapping
net use * /delete /y                             &:: remove all (fixes error 1219)

:: Server outside the domain: save the credential, prompting for the password
cmdkey /add:nas01.contoso.com /user:nas01\svc-scan /pass
cmdkey /list
net use S: \\nas01.contoso.com\scans /persistent:yes
```

WATCH OUT · NEVER PUT PASSWORDS ON THE COMMAND LINE

`net use … /user:x net use … /user:x <password> lt;password net use … /user:x <password> gt;` leaves the password in console history, process listings and EDR telemetry. Omit it and let Windows prompt, or use `cmdkey` or Credential Manager.

4.4 Reading net use errors

Error	Message (abridged)	Meaning and fix
2138	The network is not present or not started	Client broken. Steps 1–2 above.
5	Access is denied	Share or NTFS permissions. Raise to the data owner.
53	The network path was not found	Name resolution, routing, firewall or server down. `Test-NetConnection fs01 -Port 445`.
67	The network name cannot be found	Server reachable, share name wrong or removed.
85	The local device name is already in use	Letter taken (often a USB drive or stale mapping). `net use Z: /delete` or choose another letter.
86 / 1326	Password incorrect / unknown user name or bad password	Fix or remove the stored credential (`cmdkey /delete:`).

1219	Multiple connections … by the same user, using more than one user name	Disconnect all connections to that server (`net use * /delete`) and reconnect with one identity.
1272	… security policies block unauthenticated guest access	The share relies on guest access, which Windows 11 blocks. Configure the server (NAS) to require a real account.

CHAPTER 5

Method C · PowerShell

5.1 Inspect the client and its connections

INSPECT (READ-ONLY) POWERSHELL

```
Get-Service LanmanWorkstation | Select-Object Status, StartType
Get-Service LanmanWorkstation -RequiredServices
Get-SmbMapping                                  # drive letters and their status
Get-SmbConnection | Select-Object ServerName, ShareName, UserName, Dialect, Signed, Encrypted
Get-SmbClientConfiguration | Select-Object RequireSecuritySignature, EnableInsecureGuestLogons
Get-WindowsOptionalFeature -Online | Where-Object FeatureName -like 'SMB1Protocol*' | Select-Object
FeatureName, State
Test-NetConnection fs01.contoso.com -Port 445    # TcpTestSucceeded : True
```

WORKED EXAMPLE · READING THE DIALECT

`Get-SmbConnection` showing `Dialect 3.1.1` means the session uses the newest protocol with pre-authentication integrity. A value of `1.5` would reveal an SMB1 connection, which identifies exactly which server still needs upgrading before SMB1 can be removed everywhere.

5.2 Repair and re-map

ELEVATED · REPAIR (SMB1 STAYS OFF) POWERSHELL

```
Disable-WindowsOptionalFeature -Online -FeatureName SMB1Protocol -NoRestart
& sc.exe config LanmanWorkstation depend= Bowser/MRxSmb20/NSI     # sc.exe, not the sc alias
Set-Service LanmanWorkstation -StartupType Automatic
Start-Service LanmanWorkstation
Restart-Computer -Confirm
```

AS THE USER (NOT ELEVATED) · MAPPINGS POWERSHELL

```
New-SmbMapping -LocalPath 'Z:' -RemotePath '\\fs01.contoso.com\share' -Persistent $true
Remove-SmbMapping -LocalPath 'Z:' -Force
#  Alternative that also works in scripts targeting non-SMB providers
New-PSDrive -Name Z -PSProvider FileSystem -Root '\\fs01.contoso.com\share' -Persist
```

KEY CONCEPT · HARDENING THAT CAN LOOK LIKE AN OUTAGE

Windows 11 24H2 requires SMB signing on outbound connections by default and blocks insecure guest logons. Old NAS devices that cannot sign, or shares that rely on guest access, stop working after the upgrade. The right fix is on the server (enable signing, require accounts). Turning these protections off on the client with `Set-SmbClientConfiguration -RequireSecuritySignature $false` or `-EnableInsecureGuestLogons $true` is an exception-only workaround, like SMB1.

5.3 Base template · scripted client repair with evidence

REPAIR-SMBCLIENT.PS1 · BASE TEMPLATE POWERSHELL

```
<#
.SYNOPSIS  Restore the SMB client (Workstation service) without re-enabling SMB1, and record evidence.
.NOTES     Run elevated. Re-map drives afterwards in the user's (non-elevated) session.
#>
[CmdletBinding(SupportsShouldProcess)]
param(
  [string]   $TestServer = 'fs01.contoso.com',
  [string[]] $Dependencies = @('Bowser','MRxSmb20','NSI'),   # SMB1-free Windows 10/11 default
  [string]   $PackageName,                                   # display name of the package to remove
(optional)
  [string]   $EvidencePath = "$env:PUBLIC\Documents\SmbClient-$(Get-Date -f yyyyMMdd-HHmm).json"
)
$ErrorActionPreference = 'Stop'
$key = 'HKLM:\SYSTEM\CurrentControlSet\Services\LanmanWorkstation'
$ev  = [ordered]@{ Computer = $env:COMPUTERNAME; Operator = "$env:USERDOMAIN\$env:USERNAME"; StartUtc =
(Get-Date).ToUniversalTime() }

# ---- Stage 1: state before -------------------------------------------------------
$ev.ServiceBefore = (Get-Service LanmanWorkstation).Status.ToString()
$ev.DependsBefore = (Get-ItemProperty $key).DependOnService -join '/'
$ev.Smb1Before    = (Get-WindowsOptionalFeature -Online -FeatureName SMB1Protocol).State.ToString()
& reg.exe export 'HKLM\SYSTEM\CurrentControlSet\Services\LanmanWorkstation'
"$env:PUBLIC\Documents\LanmanWorkstation-backup.reg" /y | Out-Null

# ---- Stage 2: remove the package that altered SMB (if named) ---------------------
if ($PackageName) {
  $pkg = Get-Package -Name $PackageName -ErrorAction SilentlyContinue
  if ($pkg -and $PSCmdlet.ShouldProcess($PackageName, 'Uninstall-Package')) { $pkg | Uninstall-Package -
Force | Out-Null; $ev.PackageRemoved = $PackageName }
}

# ---- Stage 3: SMB1 off, dependencies corrected -----------------------------------
if ($ev.Smb1Before -ne 'Disabled' -and $PSCmdlet.ShouldProcess('SMB1Protocol', 'Disable optional
feature')) {
  Disable-WindowsOptionalFeature -Online -FeatureName SMB1Protocol -NoRestart | Out-Null
  $ev.RestartNeeded = $true
}
$want = $Dependencies -join '/'
if ($ev.DependsBefore -ne $want -and $PSCmdlet.ShouldProcess('LanmanWorkstation', "depend= $want")) {
  & sc.exe config LanmanWorkstation depend= $want | Out-Null
  $ev.DependsSetTo = $want
}
```

```
# ---- Stage 4: start and test ------------------------------------------------------
if ($PSCmdlet.ShouldProcess('LanmanWorkstation', 'Set Automatic and start')) {
  Set-Service LanmanWorkstation -StartupType Automatic
  Start-Service LanmanWorkstation -ErrorAction SilentlyContinue
}
$ev.ServiceAfter = (Get-Service LanmanWorkstation).Status.ToString()
$ev.Port445      = (Test-NetConnection $TestServer -Port 445 -WarningAction
SilentlyContinue).TcpTestSucceeded
$ev.EndUtc       = (Get-Date).ToUniversalTime()
$ev | ConvertTo-Json | Out-File $EvidencePath -Encoding utf8
Write-Host "Workstation: $($ev.ServiceAfter); 445 to $TestServer reachable: $($ev.Port445); restart
needed: $([bool]$ev.RestartNeeded)"
```

EXAMPLE · RUNNING THE TEMPLATE

INVOCATION POWERSHELL

```
.\Repair-SmbClient.ps1 -TestServer fs01.contoso.com -WhatIf
.\Repair-SmbClient.ps1 -TestServer fs01.contoso.com -PackageName '<SMB configuration package>'
```

`Get-Package` finds MSI-installed software; packages installed by other means may need their own uninstall command from the deployment tool.

CHAPTER 6

Method D · Group Policy and Intune at Scale

Once the client is healthy, the drive mappings themselves should come from a central definition, not from each user's memory. Central mappings recover automatically at the next logon and keep letters consistent.

6.1 Group Policy Preferences · Drive Maps

1. In Group Policy Management, edit a GPO linked to the users' OU and go to **User Configuration › Preferences › Windows Settings › Drive Maps**.
2. Right-click › **New › Mapped Drive**. Set **Action** to **Replace** (removes and re-creates, which also cures stale mappings) or **Update** (modifies without disconnecting).
3. Enter the **Location** (`\\fs01.contoso.com\share`), tick **Reconnect**, set a **Label as** and the drive letter.
4. On the **Common** tab enable **Item-level targeting** and target the security group that owns the data, so each team only sees its own drives.
5. On a client: `gpupdate /force`, sign out and in, and check with `gpresult /r` that the GPO applied.

6.2 Intune · script in the user's context

Intune has no native drive-map profile. Use a PowerShell **platform script** (**Devices › Scripts and remediations › Platform scripts**) with **Run this script using the logged on credentials** set to **Yes**, or a Remediation pair, which also re-checks on a schedule.

MAP-DRIVES.PS1 · RUNS AS THE SIGNED-IN USER — POWERSHELL

```powershell
$maps = @(
  @{ Letter = 'S:'; Path = '\\fs01.contoso.com\shared' },
  @{ Letter = 'T:'; Path = '\\fs01.contoso.com\team'   }
)
foreach ($m in $maps) {
  $existing = Get-SmbMapping -LocalPath $m.Letter -ErrorAction SilentlyContinue
  if ($existing -and $existing.RemotePath -ne $m.Path) { Remove-SmbMapping -LocalPath $m.Letter -Force }
  if (-not (Get-SmbMapping -LocalPath $m.Letter -ErrorAction SilentlyContinue)) {
    New-SmbMapping -LocalPath $m.Letter -RemotePath $m.Path -Persistent $true | Out-Null
  }
}
```

6.3 Intune · Remediation for the client itself (runs as SYSTEM)

DETECT-SMBCLIENT.PS1 · DETECTION — POWERSHELL

```
$deps = (Get-ItemProperty HKLM:\SYSTEM\CurrentControlSet\Services\LanmanWorkstation).DependOnService
$smb1 = (Get-WindowsOptionalFeature -Online -FeatureName SMB1Protocol).State
$bad  = ($deps -contains 'MRxSmb10' -and $smb1 -ne 'Enabled') -or ((Get-Service LanmanWorkstation).Status
-ne 'Running')
if ($bad) { Write-Output "Workstation deps: $($deps -join '/'); SMB1: $smb1"; exit 1 }
Write-Output 'SMB client healthy'; exit 0
```

REMEDIATE-SMBCLIENT.PS1 · REMEDIATION POWERSHELL

```
& sc.exe config LanmanWorkstation depend= Bowser/MRxSmb20/NSI | Out-Null
Set-Service LanmanWorkstation -StartupType Automatic
Start-Service LanmanWorkstation -ErrorAction SilentlyContinue
Write-Output "Workstation: $((Get-Service LanmanWorkstation).Status)"; exit 0
```

GOOD PRACTICE · KEEP SMB1 OFF CENTRALLY

Use the Intune settings catalog or Group Policy to keep the SMB1 client disabled and to audit for it, and fix the configuration package at source so it no longer writes an inconsistent dependency list. Prevention beats a remediation that runs forever.

CHAPTER 7

Verification, Rollback and Troubleshooting

7.1 Verification checklist

Client

- [ ] Workstation service Running, startup Automatic
- [ ] Dependencies Bowser, MRxSmb20, NSI (no MRxSmb10 unless excepted)
- [ ] SMB1Protocol feature Disabled
- [ ] Survives a restart
- [ ] No new SCM 7001 errors for Workstation

User

- [ ] `net use` shows every drive with Status OK
- [ ] `Get-SmbConnection` dialect 3.x
- [ ] Files open and save on each drive
- [ ] Drives reconnect after sign-out and sign-in
- [ ] User confirms normal operation

7.2 Rollback

Change	How to undo it
Dependencies changed	`reg import` the exported `LanmanWorkstation-backup.reg`, restart the service.
SMB1 disabled	Re-enable only under an approved exception (Method A Step 4 or the DISM command).
Package removed	Reinstall from the deployment tool only after its owner has corrected it.
Mappings changed	`net use` / `New-SmbMapping` to the previous paths; GPO mappings return at next logon.

7.3 Troubleshooting flowchart

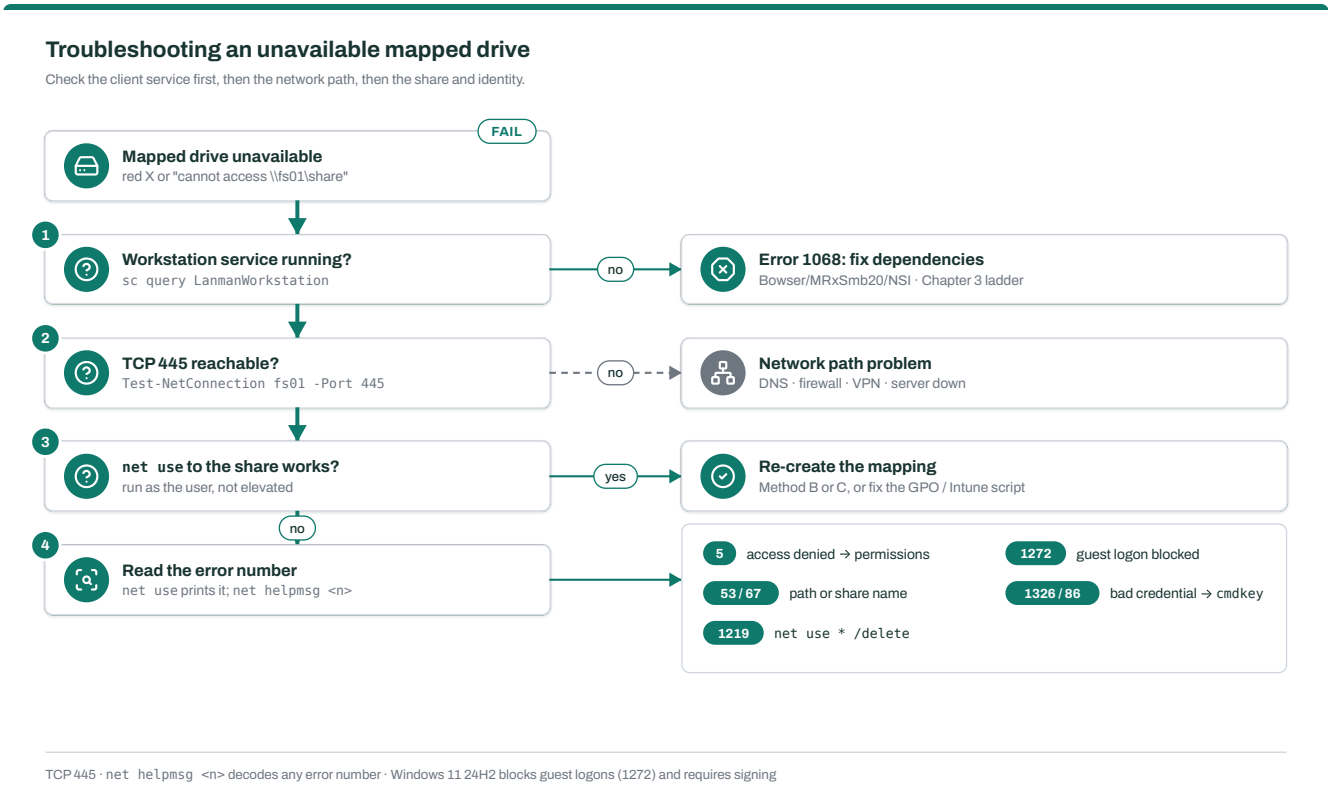


Figure 11. Rule out the client service and TCP 445 before you investigate permissions or credentials on the share.

7.4 Symptom, cause and fix

Symptom	Likely cause and fix
All drives fail at once, error 1068	Workstation dependency on a missing/disabled MRxSmb10. Correct dependencies (Method B Step 2).
Fixed, but broken again next day	Configuration package or policy reapplies the setting. Remove/correct it at source.
Drive visible in elevated prompt but not Explorer	Mapped under the elevated token. Map as the standard user; or set `EnableLinkedConnections` (DWORD 1, `HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System`) if your security baseline allows.
One NAS fails after Windows 11 24H2 upgrade	NAS does not sign SMB or uses guest access. Update its firmware/settings; enable SMB 2/3 and signing.
Red X at logon, works when opened	Drive reconnects before the network is ready. GPO *Always wait for the network at computer startup and logon*, or use GPP Replace mappings.

CHAPTER 8

Summary, Glossary and Review

8.1 Chapter summary

- The Workstation service is the SMB client. If it cannot start, every mapped drive on the PC fails.
- Error 1068 means a dependency failed. An SMB1-free Windows 11 client depends on Bowser, MRxSmb20 and NSI.
- Turning the SMB1 feature off re-registers a consistent client; turning it back on is a separate, risky choice.
- SMB1 stays off whenever the server supports SMB 2/3. Exceptions are scoped, approved and time-boxed.
- Map drives as the user, never put passwords on the command line, and manage mappings centrally with GPO or Intune.

Check your understanding

1. What is the difference between the Workstation and Server services, and which one matters for a user's mapped drives?
2. Explain why toggling SMB1 off and on fixed the reference case, and which half of that procedure actually repaired the client.
3. Write the command that sets the Workstation service's dependencies for an SMB1-free PC.
4. A user gets error 1219 mapping a second share on the same server. What happened and how do you fix it?
5. Which PowerShell command shows the SMB dialect in use, and what dialect value would reveal SMB1?

8.2 Glossary

SMB	Server Message Block, the Windows file-sharing protocol (TCP 445).
Dialect	Negotiated SMB version (1, 2.0, 2.1, 3.0, 3.0.2, 3.1.1).
LanmanWorkstation	Service name of the Workstation service, the SMB client.
MRxSmb10 / MRxSmb20	SMB1 and SMB 2/3 mini-redirector drivers.
SMB signing	Per-message integrity protection; required by default on Windows 11 24H2 clients.